Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Ungar, Eugene Ph.D.		
Project Title:	Low Power Two-Phase Active Thermal Control System		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eugene.k.ungar@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-9115
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Mail Code EC2		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2002	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	N/A		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

State-of-the-art single phase active thermal control systems (ATCSs) consume too much power to be used as spacecraft grow in size and electric power. Two-phase systems are attractive because they use the liquid/vapor phase change energy to transport heat rather than the sensible energy of single phase fluid temperature change. This decreases the required overall mass flow rate with a concomitant decrease in the pumping power requirement.
The two-phase ATCSs developed to date have suffered from relatively high power consumption, which would not allow the two-phase power gains to be fully realized on future space vehicles.

A low power two-phase ATCS (LPTPA) has been under development at NASA Johnson Space Center (JSC) for the last several years. The system differs from prior designs by not performing phase separation and using mature, efficient centrifugal pump technology. This allows a power savings of 50% over established two-phase system designs. By

Task Description:	significantly decreasing the power requirement, the LPTPA will allow significant savings in system power on future vehicles (up to 80% over state-of-the-art single phase systems). The purpose of the proposed work is to increase the LPTPA Technology Readiness Level (TRL) from 3 to 5. A prototypic LPTPA has been analyzed, designed, and was recently assembled at JSC. The prototypic LPTPA uses R134a as the working fluid, is scaled to approximate an International Space Station (ISS) sized loop, and has plumbing that minimizes gravitational effects. During the proposed effort, the LPTPA will be tested, refined, and retested. The first phase will run the LPTPA in manual control over a wide range of heat loads and operating conditions. This will prove the basic concept, demonstrate system stability, and provide information for control automation. Second, the LPTPA will be tested in automatic control mode, incorporating any refinements that are identified in the first phase of testing. The expected outcome of the project is a prototypic system whose stable operation has been demonstrated under automated control over a wide range of heat loads and operating conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completed first stage in development of a low power active thermal control system for use on physically large, high power spacecraft.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)